

Shock& Haemorrhage

By:

Dr/ Montaser Elsayy,

Professor of Cardiothoracic Surgery

Master Degree in General Surgery (MS)

MD in Cardiothoracic Surgery

Definition

- Shock exists when tissue oxygen delivery fails to meet the metabolic requirements of cells.
- *An imbalance between oxygen delivery (DO_2) and oxygen demand* can result from a global reduction in oxygen delivery, maldistribution of blood flow, impaired oxygen utilization or an increase in tissue oxygen requirements.
- Left unchecked, shock will result in *a fall in oxygen consumption (VO_2), anaerobic metabolism*, tissue acidosis and cellular dysfunction leading to *multiple organ dysfunction* and ultimately death.
- Shock is sometimes considered to be synonymous with *hypotension*.

Types of Shock

Hypovolaemic shock

This is probably the commonest and most readily corrected cause of shock encountered in surgical practice and results from a reduction in intravascular volume secondary to the **loss of blood , plasma or water and electrolytes**

Cardiogenic shock

This occurs when the heart is unable to maintain a cardiac output sufficient to meet the metabolic requirements of the body (**pump failure**) and can be caused by MI, arrhythmias, valve dysfunction, cardiac tamponade, massive pulmonary embolism, and tension pneumothorax

Neurogenic shock

This is caused by **a loss of sympathetic tone** to vascular smooth muscle.

This typically occurs following **injury to the (thoracic or cervical) spinal cord** and results in profound vasodilation, **a fall in systemic vascular resistance** and hypotension.

Causes of haemorrhagic hypovolaemic shock

Gastrointestinal haemorrhage

- Oesophageal varices
- Oesophageal mucosal (Mallory–Weiss) tear
- Gastritis
- Gastric and duodenal ulceration
- Cancer
- Diverticula

Trauma

Ruptured aneurysm

Obstetric haemorrhage

- Ruptured ectopic pregnancy
- Placenta praevia
- Placental abruption
- Post-partum haemorrhage

Pulmonary haemorrhage

- Pulmonary embolus
- Cancer
- Cavitating lung lesions e.g. TB, aspergillosis
- Vasculitis

Major blood loss during surgery

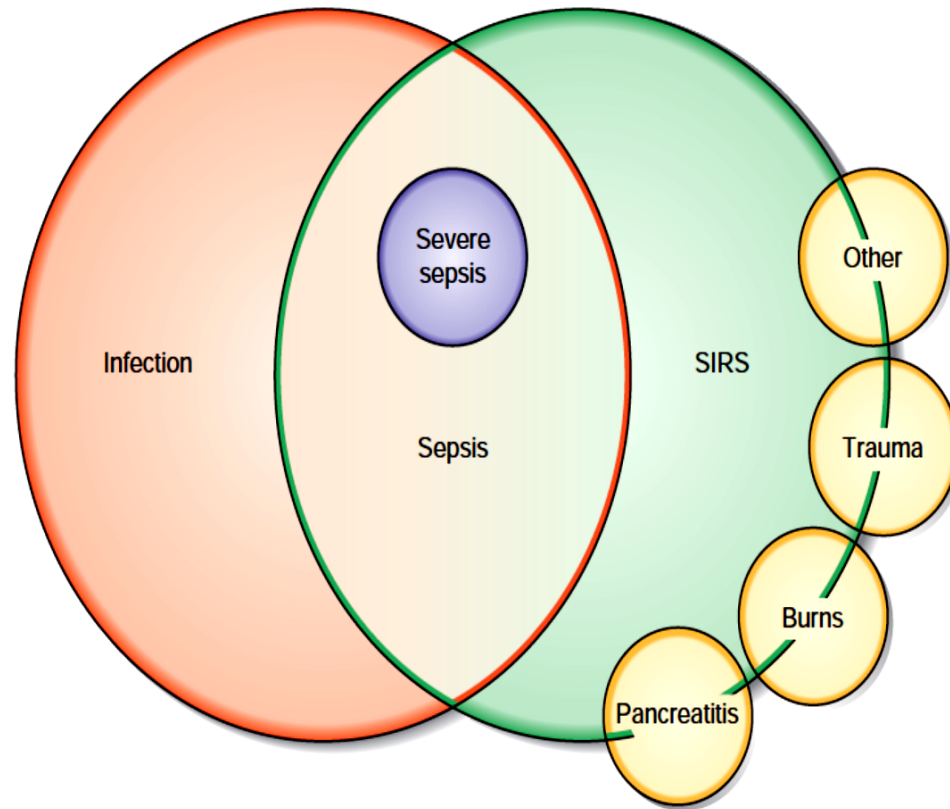
Septic shock

- Septic shock results from complex *disturbances in DO₂ delivery and VO₂* and can be defined as:

Sepsis-induced hypotension (systolic BP < 90 mmHg, mean arterial blood pressure [MAP] < 70 mmHg) and/ or tissue hypoperfusion (elevated lactate or oliguria) that persist despite adequate fluid resuscitation (~30 ml/kg).

- **Bacteria:** Sepsis usually arises from a localized infection, with Gram-ve (38%) and (increasingly) Gram +ve (52%).
- **The commonest sites** of infection leading to sepsis are **the lungs** (50–70%), **abdomen** (20–25%), **urinary tract** (7–10%) and **skin**.

Infection → Cytokine-mediated proinflammatory response → **NO SYNTHESIS** → Peripheral vasodilation (VD), **INITIAL RESPONSE**, → Redistribution of blood flow, endothelial cell activation → Endothelial cell damage → Increased vascular permeability → Capillary leak → Extravasation of fluid into the interstitial space → Haemoconcentration within the capillary → Platelet aggregation → Increase in blood viscosity → Formation of **microthrombi** within the microcirculation.,.,.,**2ry RESPONSE**,.,.,



The interrelationship between systemic inflammatory response syndrome (SIRS), sepsis and infection. Adapted from The American College of Chest Physicians and Society of Critical Care Medicine Consensus Conference Committee definitions for sepsis 1992.

Sepsis – definitions

Systemic inflammatory response (SIRS)

SIRS is defined as 2 or more of the following criteria:

- Temperature $> 38^{\circ}\text{C}$ or $< 36^{\circ}\text{C}$
- Heart rate > 90 beats per minute
- Respiratory rate > 20 per minute or $P_a\text{CO}_2 < 4.5 \text{ kPa}$
- White cell count > 12 or $< 4 \times 10^9/\text{l}$ or $> 10\%$ immature neutrophils

Bacteraemia

- The presence of viable bacteria in the blood. The presence of other pathogens in the blood is described in a similar way i.e. viraemia, fungaemia and parasitaemia.

Sepsis

- The systemic response to infection. Defined as SIRS with confirmed or presumed infection.

Severe sepsis

- Sepsis with evidence of organ dysfunction.

Septic shock

- Sepsis-induced hypotension and/or tissue hypoperfusion (e.g. oliguria, lactic acidosis) despite adequate fluid resuscitation.

Anaphylactic shock

Definition: This is a severe systemic *hypersensitivity reaction* following exposure to an allergen triggering the release of vasoactive mediators (*histamine, kinins and PG*) from basophils and mast cells.

Anaphylaxis may be:

- 1-Immunologically mediated (**allergic anaphylaxis**), when **IgE, IgG** or complement activation
- 2-Non-immunologically mediated (**non-allergic anaphylaxis**).

Common causes of anaphylaxis include:

- 1-Drugs** (e.g. NM blocking drugs, β -lactam antibiotics)
- 2-Colloid solutions** (e.g. dextrans)
- 3-Radiological** contrast media
- 4-Food stuffs** (peanuts, dairy products).

Allergen → Vasoactive mediators → VD → Intravascular volume Redistribution → Capillary leak → Reduction in cardiac output → Anaphylactic shock

Pathophysiology

A) Macrocirculation

- When assessing a patient with shock, it is useful to remember that mean arterial blood pressure (MAP) is equal to the product of cardiac output (CO) and systemic vascular resistance (SVR).

Haemodynamic and oxygen transport parameters	
$MAP = CO \times SVR$	
$DO_2 \approx CO \times 1.34 \times [Hb] \times (SaO_2 / 100)$	
$VO_2 = CO \times 1.34 \times [Hb] \times (SaO_2 - SvO_2)$	
MAP = mean arterial pressure; CO = cardiac output; SVR = systemic vascular resistance; DO_2 = oxygen delivery; [Hb] = haemoglobin concentration in g/dl; SaO_2 = arterial oxygen saturations; VO_2 = oxygen consumption; SvO_2 mixed venous oxygen saturations (sampled from pulmonary artery)	

In Hypovolaemic Shock

Hypovolaemic Shock →

1-Catecholamine release from the adrenal medulla and sympathetic nerve endings.
2-Generation of angiotensin-II (AT-II) from the renin-angiotensin system.



Tachycardia + Increased myocardial contractility act to *preserve COP + VC*

1-The resulting tachycardia and increased myocardial contractility act to *preserve COP*

2-VC acts to maintain BP and divert the available blood to vital organs (e.g. brain, heart and muscle) and away from non-vital organs (e.g. skin and gut).

In septic shock

- The (initial) cardiovascular response is a **reflex tachycardia** and an increase in SV resulting in an **increased COP**.
- Clinically this manifests as warm, well-perfused peripheries, a low diastolic BP and **raised pulse pressure**.

DO₂ and VO₂ is compromised at the cellular level.

However, as septic shock progresses *endothelial dysfunction* results in significant **extravasation of fluid** and a loss of intravascular volume.

Ventricular dysfunction also impairs the compensatory increase in COP.

As a result, peripheral perfusion falls and the clinical signs may become indistinguishable from those associated with the low-output state described above

In Neurogenic Shock



Traumatic disruption of sympathetic efferent nerve fibres results in loss of vasomotor tone, peripheral **VD** and a fall in **SVR**.



Loss of cardiac accelerator fibres (T1–4) and anhidrosis as a result of loss of sweat gland innervation also frequently occur, with patients typically presenting with *hypotension, bradycardia and warm, dry peripheries*.

In Cardiogenic Shock

Typically presents with signs of *a low output state* although, unlike hypovolaemic shock, circulating volume is typically normal or increased with increased circulating AT-II and aldosterone.

If associated with left ventricular failure, there may be pulmonary oedema.

B) Cellular Function

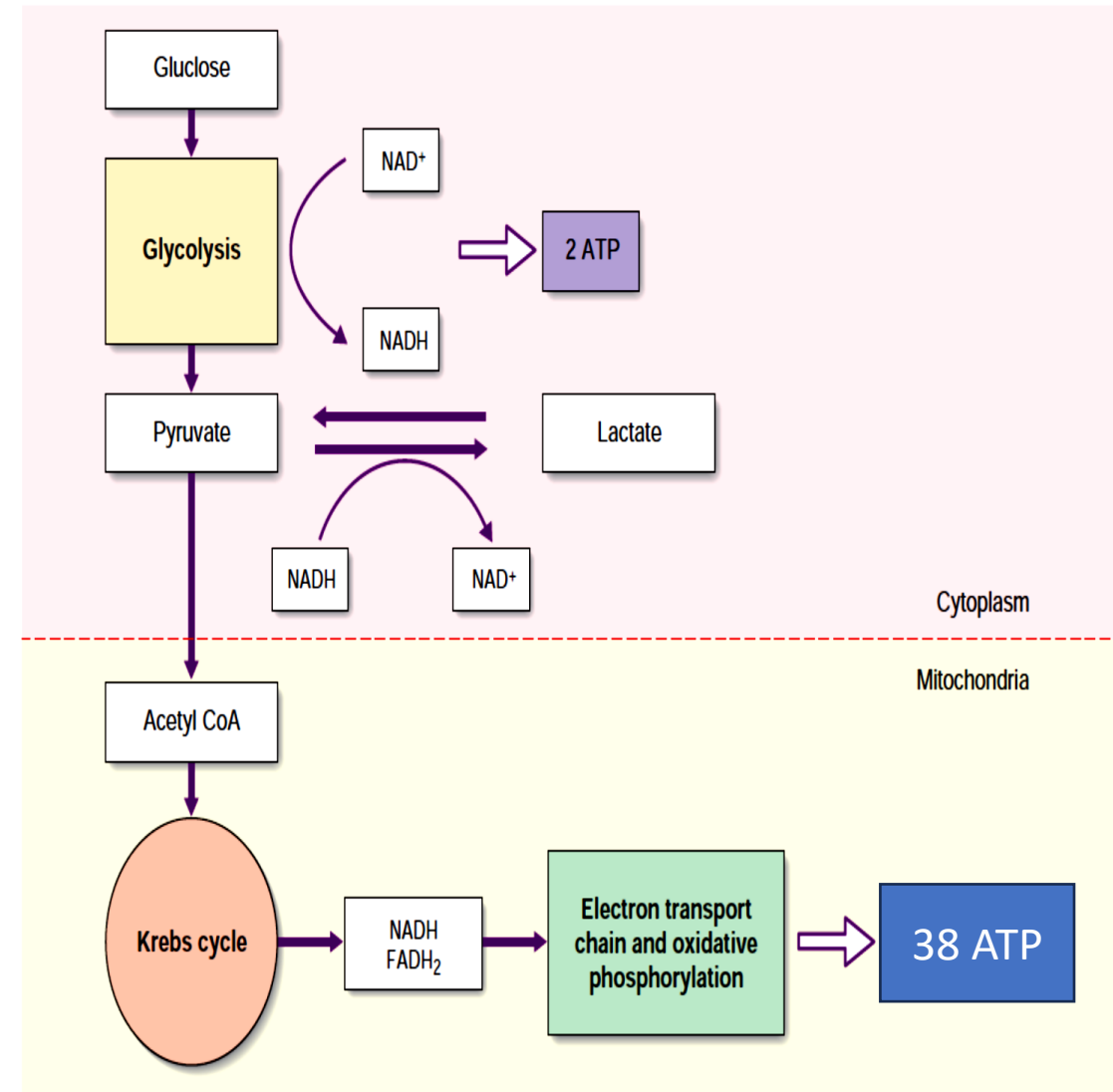
Glycolysis (ATP Synthesis from glucose)

Aerobic Pathway

Under normal (aerobic) conditions, glycolysis converts glucose to pyruvate which is converted to acetyl-coenzyme-A (acetyl-CoA) and enters the Krebs cycle.

Oxidation of acetyl-CoA in the Krebs cycle generates nicotinamide adenine Dinucleotide (NADH) and flavine adenine dinucleotide (FADH₂), which enter the electron transport chain and are oxidized to NAD⁺ in the oxidative phosphorylation of adenosine diphosphate (ADP) to ATP.

The oxidative metabolism of glucose is energy efficient, yielding up to 38 moles of ATP for each mole of glucose, but requires a continuous supply of oxygen to the cell.



Anaerobic Pathway

Hypoxaemia blocks mitochondrial oxidative phosphorylation, inhibiting ATP synthesis.

This leads to a decrease in the intracellular ATP/ADP ratio, an increase in the NADH/ NAD⁺ ratio and an accumulation of pyruvate that is unable to enter the Krebs cycle.

The cytosolic conversion of pyruvate to lactate allows the regeneration of some NAD⁺, enabling the limited production of ATP by anaerobic glycolysis. However, anaerobic glycolysis is significantly less efficient, generating only 2 moles of ATP per mole of glucose and predisposing cells to ATP depletion

The effect (Complications) of shock on individual organ systems

Clinical effects of shock	
Nervous system	• Restlessness, confusion, stupor, coma • Encephalopathy and/or delirium common in sepsis
Renal	• Renal hypoperfusion → activation of rennin–angiotensin system • Oliguria (< 0.5 ml/kg/h urine) → anuria • Acute renal failure → ↑ urea, ↑ creatinine, ↑ K⁺ & metabolic acidosis
Respiratory	• Tachypnoea • ↑ Ventilation/perfusion (V/Q) mismatch & ↑ shunt → hypoxia • Pulmonary oedema (common in cardiogenic shock) → hypoxia • Acute lung injury and acute respiratory distress syndrome → hypoxia
Cardiovascular	• ↓ Diastolic pressure → ↓ coronary blood flow • ↓ Myocardial oxygen delivery → myocardial ischaemia → ↓ contractility & ↓ CO • Acidosis, electrolyte disturbances and hypoxia predispose to arrhythmias • Widespread endothelial cell activation → microcirculatory dysfunction
Gastrointestinal	• Splanchnic hypoperfusion → breakdown of gut mucosal barrier • Stress ulceration • Translocation of bacteria/bacterial wall contents into blood stream → SIRS • Acute ischaemic hepatitis.

Clinical assessment of shock

Conscious level	Restlessness, anxiety, stupor and coma are common features and suggest cerebral hypoperfusion
Pulse 	Low volume, thready pulse consistent with low-output state; high volume, bounding pulse with high-output state
Blood pressure 	Changes in diastolic may precede a fall in systolic blood pressure, with ↓ diastolic in sepsis and ↑ in hypovolaemic and cardiogenic shock
Peripheral perfusion	Cold peripheries suggest vasoconstriction (↑ SVR); warm peripheries suggest vasodilation (↓ SVR)
Pulse oximetry	Hypoxemia common association of all forms of shock and ↓ tissue O ₂ delivery
ECG monitoring	Myocardial ischaemia commonest cause of cardiogenic shock but common in all forms of shock
Urine output	< 0.5 ml/kg/h suggestive of renal hypoperfusion
CVP measurement	Low CVP with collapsing central veins consistent with hypovolaemia
Arterial blood gas	Metabolic acidosis and ↑ lactate consistent with tissue hypoperfusion

Management



General principles

The management of shock is based upon the following principles:

- 1- Identification and treatment of the underlying cause*
- 2- The maintenance of adequate tissue oxygen delivery.*
- 3-As with most clinical emergencies, treatment and diagnosis should occur simultaneously with the immediate assessment and management following an **Airway, Breathing, Circulation (ABC)** approach.
- 4- The early recognition and treatment of potentially reversible causes (e.g. bleeding, intra-abdominal sepsis, myocardial ischaemia, pulmonary embolus, cardiac tamponade) is essential and may be facilitated thorough clinical examination (Table) and focused investigations.
- 5-Most patients with shock will require admission to *(ICU)*.

Airway Management

OXYGEN:

Hypoxaemia must be prevented and, if present, rapidly corrected by maintaining a clear airway (e.g. head tilt, chin lift) and administering high flow oxygen.

ABG Monitoring:

The adequacy of this therapy can be estimated continuously using pulse-oximetry (SpO_2), but frequent arterial blood gas analysis allows a more accurate assessment of oxygenation ($P_a.O_2$), ventilation ($P_a.CO_2$) and indirect measures of tissue perfusion (pH, base excess, HCO_3 and lactate).

MV:

In patients with severe hypoxaemia, cardiovascular instability, depressed conscious level or exhaustion, intubation and ventilatory support may be required.

Circulatory Support

FLUIDS

- Initial resuscitation should be targeted at arresting haemorrhage and providing fluid (*crystalloid or colloid*) to restore intravascular volume and optimize cardiac preload.
- It is common practice to use blood to maintain a haemoglobin concentration $> 10 \text{ g/dl}$ during the initial resuscitation of shock if there is evidence of low central venous saturations (*measured from a central venous catheter*).

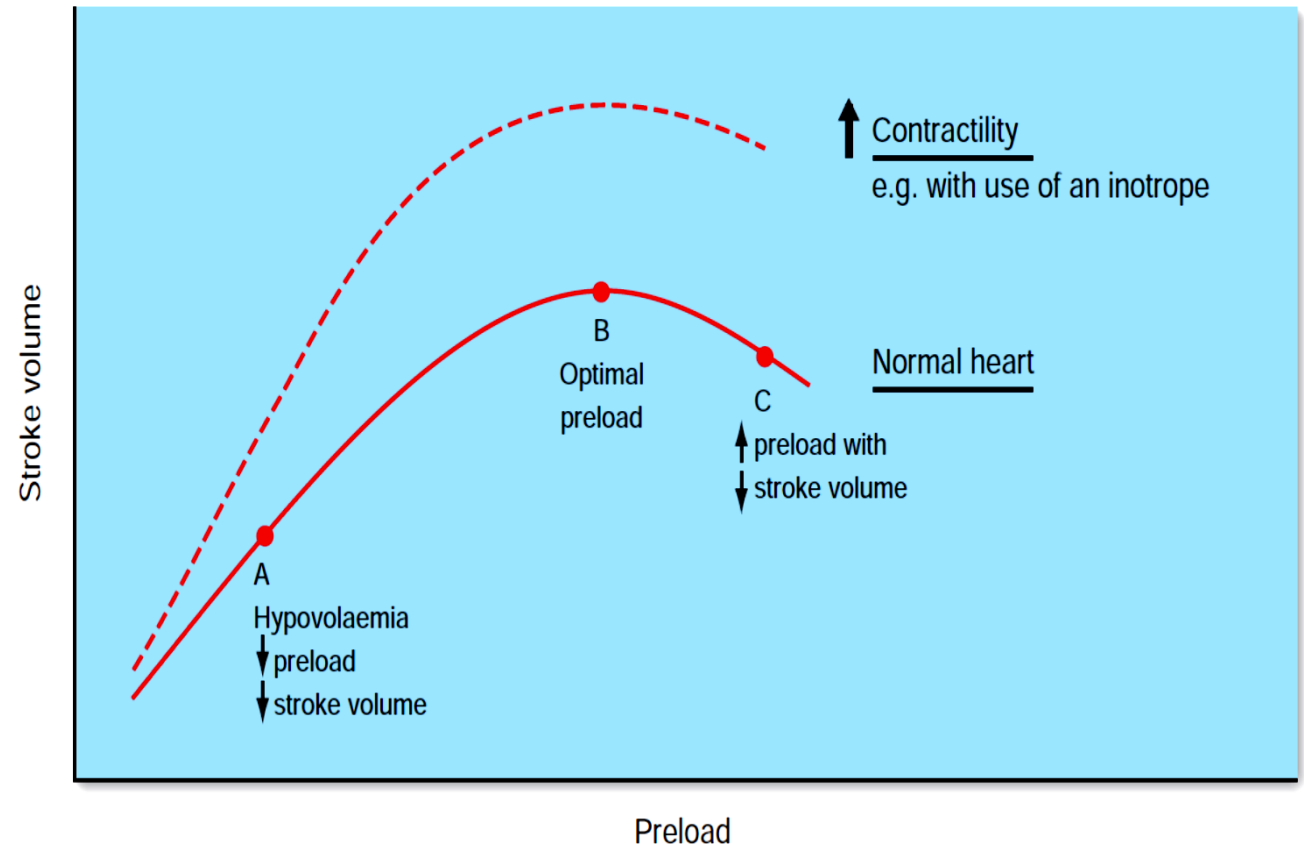
‘Fluid responsiveness’

↓ Tachycardia, ↑ BP, and improving peripheral perfusion and urine output in response to a series of $250\text{--}500 \text{ ml}$ fluid challenges indicate ‘fluid responsiveness’ and suggest that further fluid and optimization of preload may be required.

Inotropes and/or Vasopressors

Once parameters stop improving it is unlikely that further fluid will be beneficial, particularly if there is an associated fall in oxygen saturation and the development of pulmonary oedema.

- As more invasive monitoring allows the acid–base status, central venous pressure (CVP), pulmonary artery wedge pressure (PAWP), CO and mixed (SvO₂) or central to be used to further assess the response to fluid (Fig).
- If blood pressure remains low and/or signs of inadequate tissue oxygen delivery persist despite fluid resuscitation and the optimization of preload, then inotropes and/or vasopressors may be required.
- **Vasopressors** (e.g. noradrenaline) cause peripheral vasoconstriction and an increase SVR.
- **Inotropes** (e.g. dobutamine) increase myocardial contractility, stroke volume and cardiac output.



Frank–Starling curve. Demonstrating the relationship between ventricular preload and stroke volume.

Choice of Inotrope or Vasopressor

- The initial choice of inotrope or vasopressor therefore depends upon the underlying aetiology of shock and an understanding of the main physiological derangements (*Table*).
- Adrenaline**, which has both vasopressor and inotropic effects, is a useful first line drug in the emergency treatment of shock.

Effects of commonly used vasoactive drugs			
Drug	CO	SVR	Main effects
★ Adrenaline	↑	↑	α- & β-agonist; positive inotrope and vasopressor
★ Noradrenaline	↔/↓	↑	α-agonist; vasopressor
★ Dobutamine	↑	↓	β ₁ -agonist; positive inotrope and systemic vasodilator
Dopamine	↑	↓	β ₁ -agonist (at doses > 5 µg/kg/min); positive inotrope and systemic vasodilator
Dopexamine	↑	↓	β ₁ -agonist; positive inotrope and systemic vasodilator
Levosimendan	↑	↓	Calcium sensitizer; positive inotrope and systemic vasodilator
Milrinone	↑	↓	Phosphodiesterase inhibitor; positive inotrope and systemic vasodilator
★ Glyceryl trinitrate	↑	↓	Nitric oxide-mediated vasodilatation
CO = cardiac output; SVR = systemic vascular resistance			

Management of Hypovolaemic shock

- The commonest cause of acute hypovolaemic shock in surgical practice is **bleeding due to trauma, ruptured aortic aneurysm**, obstetric haemorrhage.
- Normal adult blood volume is about **7%** of body weight, with a 70 kg man having an estimated blood volume (EBV) of around **5000 ml**.
- The severity of haemorrhagic shock is frequently classified according to percentage of EBV lost where class I (**< 15%**) represents a compensated state (as may occur following the donation of a unit of blood) and class IV (**> 40%**) is immediately life threatening (Table).
- The term '**Massive Haemorrhage**' has a number of definitions including:

Loss of EBV in 24 hours;

Loss of 50% EBV in 3 hours

Blood loss at a rate ≥ 150 ml/min.

- Arrest of haemorrhage and intravascular fluid resuscitation should occur **concurrently**; *there is little role for inotropes or vasopressors in the treatment of a hypotensive hypovolaemic patient.*

Estimated blood loss and presentation of hypovolaemic shock				
	★ Class I	Class II	Class III	★ Class IV
Blood loss (ml)	< 750	750–1500	1500–2000	> 2000
Blood loss (% EBV)	< 15	15–30%	30–40%	> 40%
Pulse	< 100	> 100	> 120	> 140
Blood pressure	Normal	Decreased	Decreased	Decreased
Respiratory rate	14–20	20–30	30–40	> 35
Urine output (ml/h)	> 30	20–30	5–15	Negligible
CNS symptoms	Normal	Anxious	Confused	Lethargic
Adapted from Committee on Trauma: Advanced Trauma Life Support Manual				

Types of Fluids

- The type of fluid used (**crystalloid or colloid**) is probably less important than the adequate restoration of circulating volume itself.
- In the case of life-threatening or continued haemorrhage, **blood** will be required early in the resuscitation.
- Ideally, fully cross-matched packed red blood cells (PRBCs) should be administered, but type-specific or O Rhesus-negative blood may be used until it becomes available.
- A haemoglobin concentration of **7–9 g/dl may be sufficient** to ensure adequate tissue oxygen delivery in stable (non-bleeding) patients, but a haemoglobin target of > 10 g/dl may be more appropriate in actively bleeding patients.
- **Massive transfusion** can lead to hypothermia, hypocalcaemia, hyper- or hypokalaemia and coagulopathy.



Coagulopathy

- **The acute coagulopathy of trauma (ACoT)** is well recognized and multifactorial:

***Dilution** of clotting factors and platelets as a result of fluid resuscitation, combined with their **consumption** at the point of bleeding, results in clotting factor deficiency, thrombocytopaenia and coagulopathy.*

- **Hypothermia, metabolic acidosis and hypocalcaemia** also significantly impair normal coagulation.
- Resuscitation strategies aggressively targeting the 'lethal triad' of hypothermia, acidosis and coagulopathy appear to significantly improve outcome.

Correction of Coagulopathy

- Where possible, correction of coagulopathy should be guided by laboratory results (platelet count, prothrombin time, activated partial thromboplastin time and fibrinogen concentration).
- **Clotting factor deficiency** is normally treated by the administration of **fresh frozen plasma (FFP)** (10–15 ml/ kg)
- **Fibrinogen deficiency** (< 1.0 g/l) is best treated with fresh frozen plasma or cryoprecipitate (usually one ‘pool’ of 10 single donor units).
- **Thrombocytopenia** or platelet dysfunction by the administration of **platelets** (usually one ‘pool’ or adult dose containing $2-3 \times 10^{11}$ platelets).
- **The antifibrinolytic, tranexamic acid**, can be used to inhibit fibrinolysis and has been shown to reduce mortality from bleeding when used early (< 3 hours) and empirically following major trauma.

Correction of Coagulopathy

- In the case of **rapid haemorrhage**, it is often not possible to use traditional laboratory results to guide the correction of coagulopathy because of the time delay in obtaining these results.
- This has led to a formula-driven approach to the use of **P-RBC, FFP and platelets** targeting the early empirical treatment of coagulopathy.
- Administration of **warmed PRBC and fresh frozen plasma (FFP) in a 1:1 ratio** as soon as possible in the resuscitation of major haemorrhage following trauma in conjunction with platelet transfusions to maintain platelets $> 100 \times 10^9$.
- A recombinant form of activated factor VII (rVIIa) is approved for the management of bleeding in haemophiliacs with inhibitory antibodies to factors VIII or IX.

Thank



You